# Developing an Architecture for IoT Interoperability in Healthcare: A Case Study of Real-time SpO2 Signal Monitoring and Analysis

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***Abstract*—The article presents a novel approach to address the interoperability issues in the Internet of Things (IoT) systems. Based on the IoT Reference Architecture, an IoT system architecture integrated with gateways and data converter solutions has been proposed to enhance the information exchange and data sharing across interconnected devices, networks and platforms.**

**In addition, distributed open-source analytic platforms for real-time computational and streaming processing has been introduced to the proposed architecture. A case study of real-time oxygen saturation (SpO2) monitoring and analysis has been implemented to demonstrate the feasibility of the proposed architecture. The experimental results show the efficiency in terms of collecting and analyzing large volumes of data.**

***Index Terms*—Internet of Things, Interoperability, Industrial Internet Reference Architecture, Distributed Big Data Analytics, Apache Spark.**

## I. Introduction

The Internet of Things (IoT) has been widely applicable in a variety of fields, including healthcare. The emergence of IoT-based healthcare systems has provided unprecedented benefits that would enhance the quality and efficiency of the treatments and accordingly improve the health of the patients [1]. Furthermore, the large amount of data generated by IoT devices has created many research challenges and opportunities. Therefore, enterprises and health institutions must create a system where they can collect, manage, and analyze sensor data in a scalable and cost-effective manner [2].

Although the healthcare sector is rapidly adopting various IoT solutions creating a huge network of medical things, the interoperability issues remain a hindrance for the widespread adoption of health systems. As the IoT system is a network composed of heterogeneous protocols, standards, and platforms supported by different manufacturers, requirements and usage intentions, IoT systems may be unable to simultaneously support multiple protocols or standards in an extensible way. This lack of interoperability results in serious issues related to confidentiality, reliability, transparency and securability [3]–[5]. There have been very few studies dedicated on addressing this issue in IoT-based healthcare systems [6]. Moreover, there has been an explosion of companies and solutions into the market, creating a very fragmented landscape within the IoT architectures [7]. As a result, this fragmentation has been one of the major obstacles for the wide implementations of IoT technologies and applications. Hence, standardized structures, IoT Reference Architectures [8], have been initiated to define the rules of behavior and functioning of the elements within the IoT ecosystem. Several IoT Reference Architectures have been proposed in order to standardize concepts and implementation of IoT systems in different domains such as Industrial Internet Reference Architecture (IIRA) [9], The WSO2 IoT Reference Architecture [10] and SAT IoT [11].

In addition, the power of artificial intelligence and machine learning have been adopted in the IoT systems in order to pursue real-time services and accurate analysis [12]–[14]. In this context, leveraging big data platforms that can assist in consuming and reading diverse data sources as well as in accelerating the data integration process becomes vital. On one


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hand, it opens approaches in big data processing to transform a huge volume of sensor-collected data into valuable insights. On the other hand, it raises a big issue on how to integrate data analysis tools into the IoT systems and process a huge amount of data in real-time [15], [16]. In fact, solving this question is hard because of existing difficulties in streaming a large amount of data through networks and limitations in big data processing. Therefore, having the ability of distributed computing along with salable event streaming in IoT systems is a key factor.

**The main focus of this work are as follows:**

- Providing a holistic view on existing state-of-the-art and challenges for the IoT architecture in healthcare.
- Providing strategies to design and develop a skeletal IoT infrastructure for smart connected health applications.
- Proposing the IoT gateway, data converter and data analytic integration to enhance the IoT interoperability and real-time big data analysis.
- Investigating a case study concerning the development of a small-scale laboratory for smart healthcare.

The rest of this article is organized as follows: In Section II, we review the backgrounds and related works. Then, we present our proposed ecosystem based on the IIRA and the solutions for addressing the interoperability issues and integrating data analysis platforms in Section III. The case study and experimental results are provided in Section IV. Finally, conclusions and future works are presented in Section V.

## II. Background and Related Works

IoT has undoubtedly re-calibrated the healthcare industry by redefining the space of devices and people interactions in delivering healthcare solutions. This session discusses the existing state-of-the-art and primary issues of IoT systems in the healthcare sector in a systematic way.

### A. IoT Reference Architectures

The healthcare system is complex with thousands of heterogeneous sensors connected through the Internet, so there is a crucial need for a flexible architecture. According to recent research [17], [18], the basic architecture widely used in healthcare consists of three layers, including the application, network, and perception layers. There are some extensions that add more abstractions such as middle-ware, five-layer and service-oriented architectures. However, these architectures have their own disadvantages in terms of compatibility, security and scalability [19]–[21]. There are several initiatives and attempts to define reference architectures, that will standardize the architectural design of IoT systems. The ultimate goal of referencing architectures is to reach a higher degree of scalability, end-to-end security, automation operations and achieving interoperability both in conceptual and practical terms, as this increases connectivity within the IoT ecosystem thus ensuring overall project success.

Among proposed reference architectures, IIRA, developed by the Industrial Internet Consortium (IIC), can cover

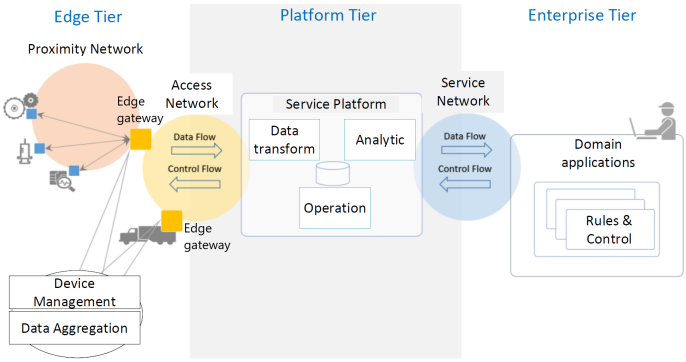


Fig. 1. Three-Tier IIoT system architecture. (Adapted from Lin et al. [9])

a broader range of possible application sectors including healthcare. Regarding the model pattern, IIRA is based on the three-tier industrial IoT architecture but more advanced improvements added, comprising the Edge, Platform, and Enterprise tiers, and the networks connecting them, as presented in Fig. 1. The description of IIRA is illustrated below:

- The Edge tier collects data from the Edge nodes over a local network (Proximity Network). This Proximity Network connects machines, actuators, sensors and other intelligent assets. It combines these edge nodes into one or more clusters, which are then linked to the Access Network via Edge Gateways. The Edge Gateways often act as an access point for device management and is able to pre-aggregate or otherwise pre-process selected data. The Access Network provides connectivity for data and control flows between the Edge and Platform tier. It could be a closed corporate network or a Virtual Privacy Network. For data transmission, specialized IoT protocols such as Message Queuing Telemetry Transport (MQTT) or Constrained Application Protocol (CoAP), but also specific REST APIs can be used.
- The Platform tier essentially consists of a non-domain-specific service platform that provides services for transforming, storing and analyzing the data as well as supporting functions for operating the platform including device management. In addition, the Platform-Tier is also the central data hub. It processes and consolidates data flows from the Edge Tier and forwards them to the Enterprise Tier as needed. Conversely, it receives and processes control commands from the Enterprise Tier and forwards them to the Edge Tier via the Service Network which facilitates the connection between the services of the Platform Tier and the Enterprise Tier.
- The Enterprise Tier receives data from Platform and Edge Tiers and sends control commands to Platform and Edge Tiers. This tier contains all higher-level business rules and control mechanisms. In reality, these are domain-specific business applications such as ERP, CRM or decision support systems, but also user interfaces for platform operations.


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To get a complete picture of various IoT architectures used in the healthcare sector, we prescribe related studies in different main application areas.

*1) Mobile Health:* Providing data security and privacy solutions is needed to maintain patient confidentiality. A privacy preservation framework proposed by Ukil et al. [22] provided a negotiation based architecture to find a solution for utility-privacy trade-off in e-Health data management. In this work, they did not address the security issue when deploying the negotiation framework.

*2) Home Healthcare:* Yang et al. [23] proposed an iHome Health-IoT platform for healthcare services at home, which consists of a smart medical service layer; a medical resource management layer; and a sensor data collecting layer. The platform is developed based on an intelligent medicine box (iMedBox) [24] with enhanced connectivity and interchangeability for the integration of devices and services. However, the connection between layers is limited to some predetermined communication protocols. The monitoring system is not compatible for long-term usage due to the physical limitation of large size and rigid package.

Ray et al. [25] presented a 5-layered system named Home Health Hub Internet of Things (H3IoT) to disseminate the healthcare of elderly people at home. However, this model does not fit to build scale applications due to lack of the capability to handle the growing amount of sensor data.

*3) Hospital Management:* Yu et al. [26] used a three-layer IoT architecture to implement a smart hospital platform for improving efficacy of present hospital information systems, such as: fixed information point, inflexible networking mode, and related parameters. Another Automating Design Methodology (ADM) system for smart rehabilitation of old age population is devised by Fan et al. [27]. Such an ontology based platform creates a rehabilitation strategy and re-configures the medical resources according to patients' specific requirements quickly and automatically. However, both studies are not able to address the need of data analysis after having patient data.

In general, IoT-based systems implemented are typically represented in a bottom–up approach, where the sensors and network gateways are in the lower layers, and the services and applications are in higher layers [28], [29]. The IoT domain is like vertical silos of various IoT applications without horizontal inter-connectivity between them [30]. Currently, lack of interoperability is a main issue that influences wide adoption and acceptability of IoT application. Based on our investigation, not many projects have addressed issues related interoperability and real-time data analytic in the healthcare domain.

### *B. Main Challenges of Building IoT Systems in Healthcare*

After showing the basic background and related works toward IoT Architecture. Next, we discuss the main challenges that we want to address for removing the hindrances of widespread IoT development in healthcare.

**Challenge 1: IoT Interoperability**

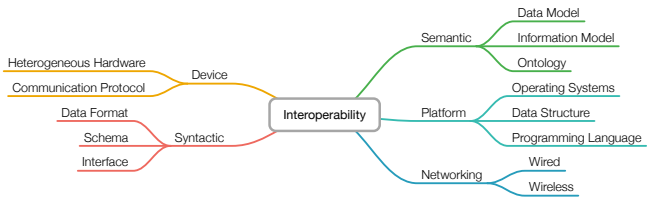


Fig. 2. Interoperability issues in IoT. (Adapted from Noura et al. [31])

Interoperability is defined as "The ability of two or more components to share information and use information exchanged" [32]. Research studies addressing interoperability issues in healthcare are still limited. In the healthcare domain, different vendors offer various products, devices and protocols. Usually, they have no obligation to follow the rules. This continuing process of mutation in terms of protocols and standards raises a number of interoperability issues. Hence, it is important to consider a broad range of topics such as devices, networking, applications, syntactic and semantic data for standardization purposes.

In order to understand IoT interoperability, we need to take an approach to classify it. As indicated in Fig. 2 IoT interoperability can be explained from different aspects such as networking interoperability, devices interoperability, platform interoperability, syntactic interoperability and semantic interoperability that we examine them as follows.

*1) Devices interoperability:* The devices interoperability issue is caused by two reasons. First, IoT devices are composed of heterogeneous hardware. The microcontroller architecture and core system characteristics of those IoT devices such as RAM, processor speed and communication technologies vary widely between various brands and models, so that they have different ways for communication. Second, diversification of connectivity protocols depends on different requirements of the IoT market. For example, smart IoT devices are typically built on existing standards like the well-known Wi-Fi and Bluetooth, but in some domains, non-standard protocols commonly used such as ANT+ standard in mobile health [33], ZigBee [34] or LoRa and Sigfox [35] in environmental monitoring systems. Therefore, devices interoperability is concerned with (1) the exchange of information between heterogeneous devices and communication protocols; (2) the ability to integrate new devices into the IoT platform.

*2) Network interoperability:* Network interoperability is the continuous ability to send and receive data within interconnected networks, maintaining the level of quality required by the end user without any negative effects [31]. In other words, network interoperability is crucial in order to achieve end-to-end communication. The more complex networks exist, the greater the need to ensure that they can be inter-operable to enable end-to-end communication possible.

To make systems interoperable, each system should be able to exchange messages with other systems through various types of networks. For many reasons, effective


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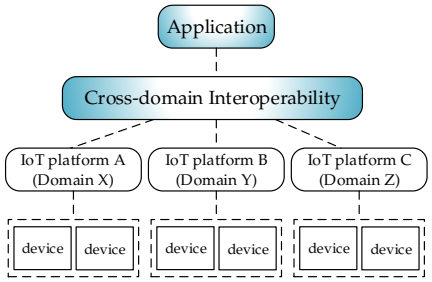


Fig. 3. Cross-domain interoperability. (Adapted from Noura et al. [31])

implementation of network interoperability is considered a challenging job. The key point is the correct balance between the telecommunications provider's liabilities and the associated benefits. Basically, designing the network architecture for interoperability implies the willingness to accept a complex set of benefits and associated liabilities.

*3) Platform interoperability:* The problem of platform interoperability in IoT arises due to the availability of diverse data structures, operating systems (OSs), programming languages and access mechanisms for things and data. This non-uniformity causes hindrance to develop cross-platform and cross-domain IoT applications [36]. Specifically, application developers need to acquire comprehensive knowledge of various APIs and information models of each platform to be able to adapt their applications from one platform to another. Fig. 3 shows that IoT cross-platform applications can access external IoT platforms for exchanging data at interoperability level. Considering the following scenario: a patient uses an IoT cross-platform to connect to a smart health platform to continuously track his health conditions (heart rate, oxygen saturation, and glucose level) and, in an emergency, the platform locates him and sends alerts to an ambulance. The cross-platform is also capable of accessing a smart-city platform to buy a ticket to the desired destination and displays the shortest path to the bus/train station. The cross-platform interoperability between things and data in this scenario allows interoperability across separated IoT platforms related to one vertical domain such as smart home, smart healthcare, smart garden, etc.

*4) Syntactic interoperability:* Syntactic interoperability refers to inter-operation of the format as well as the data structure used in any exchanged information between heterogeneous IoT system entities [31]. This issue occurs when the sender's encoding rules are inconsistent with the receiver's decoding rules, which leads to mismatching message parse trees. There are several existing approaches to enhance syntactic interoperability. For example, methods of mash-up devices with service-oriented computing (SOC)-based architecture [37], RESTful Web Services [38], Open Standard Protocols [39], and Closed Protocols [40]. They are all designed for remote access to entities for exchanging information. Until now, a vast range of open and proprietary wireless protocols and technologies have arisen for syntactic interoperability, such as Wireless HART [41], IEEE 802.15.4 standards [39], as well as proprietary technologies such as Z-Wave [42]. However, these standards are heterogeneous and incompatible with each other. Thus syntactic interoperability remains a challenge.

*5) Semantic interoperability:* To maximize the ability of information shared by IoT systems to be interpreted in the same way, a level of Semantic Interoperability is required [43]. Then, an identical syntactic representation is also considered to support semantic interoperability, [44]. To be more precise, the data generated by things may have a defined data format (e.g. JSON, XML or CSV), but the data structures and schemas used by different systems are usually dissimilar and not always compatible. Let's consider an example in heathcare domain, an e-Health platform receives a medical report with one of the patient's conditions being listed as "Grinch Syndrome". The platform cannot accurately map "Grinch Syndrome" to any other disease, since it is likely that most clinicians would prefer to use more clinical terminologies. The name sounds pretty innocuous, so it may not be that important to map, right? Wrong. "Grinch Syndrome" is just another way of saying "Postural Orthostatic Tachycardia Syndrome", meaning that the patient's heart is disproportionately small, sometimes "two times too small,". Either name indicates a very serious ailment, and if the platform were to leave it out of the EHR or EMR due to the inability to recognize the name, physicians would not have access to vital patient information.

Possibilities for errors like these necessitate a common language between healthcare systems, but all using the same notation such as common mapping dictionaries or data structures in health records is not realistic in practice. Therefore, developing an integrated semantic function converting exchanged data structures to the one that can be consumed is indispensable.

**Challenge 2: Real-time Big Data Analysis**

Implementing the real-time big data analysis function is a huge task when building a smart healthcare system. Real-time analysis can help doctors reach insight faster and handle streaming data sources that give their analysis more depth. Thus, the system should be able to deal with spikes in data volume at high speed and be able to scale up as data grows. However, processing the whole data on single machine or monolithic architecture is not practical due to computational limitations, reliability, scalability and power consumption concerns. This is alleviated by using Kafka and Spark clusters [45], which are powerful distributed computing frameworks.

Apache Kafka[1] is a distributed streaming platform for creating real-time data pipelines. It is scalable, fault-tolerant and very fast, which is a perfect match for big data applications.

Apache Spark[2] is a data processing framework that can quickly perform tasks on very large data sets, and can also

[1]https://kafka.apache.org
[2]https://spark.apache.org


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TABLE I
OPEN-SOURCE IOT PLATFORMS COMPARISON

| Platform | Architecture | Protocols | Device Management | Security | Analysis | Visualization | Database |
|---|---|---|---|---|---|---|---|
| ThingsBoard | Monolithic, Micro-services | HTPP(s), REST, MQTT, CoAP, ModBus, OPC-UA, BLE, CAN | Advanced | MQTT/HTTP(s) encryption, Device authentication, Credentials management | Connection with data analysis platforms and tools | Advanced | Built-in SQL, NoSQL, Hybrid |
| DeviceHive | Monolithic | HTTP, MQTT, REST | Basic | Device authentication | No | Basic | NoSQL plugin |
| Zetta | Monolithic | REST, HTPP | No | No | Yes | Basic | Need plugins |
| SiteWhere | Monolithic, Micro-services | HTTP, MQTT, AMQP, Stomp, CoAp | Protocols | HTPP(s), Spring Security | Security | No | SQL, NoSQL plugin |

distribute data processing tasks across multiple clusters, either on its own or in tandem with other distributed computing tools.

## III. PROPOSED SYSTEM ARCHITECTURE

In this session, we propose a smart system based on IIRA model and ThingsBoard platform. It can be used as a skeletal IoT Architecture enduring basic requirements such as device management, security and scalability. The system is capable of:

- Providing a strategy of using gateways, integration and data converter supported as build-in functions in ThingsBoard to address the IoT Interoperability mentioned in the previous session.
- Integrating a real-time data analysis with Kafka and Spark platforms for big data analytic applications.

### A. Choosing Open-source IoT Platforms

To reduce costs and the time taken to complete building an efficient system, utilizing an open-source IoT platform is probably a good approach. However, the selection of an open-source IoT platform is not an easy task. It depends on which applications and varies according to the context and platform providers. In addition, IoT platforms must share some common characteristics like managing and interconnecting the IoT endpoints, collecting and processing data, along with providing data visualization and IoT application development tools.

There are several surveys that review existing IoT platforms [46], [47]. Most of them solved performance issues [48], [49], while others specialize in specific domains such as smart building [50] or agricultural [51]. For the scope of this work, we focus on engineering aspects to enhance interoperability and the capability of developing interfaces leveraging Kafka and Spark platforms for big data processing, and also ensure other attributes like how easy to customize and configure the open-source platform, the ability of devices management and security. Considering these requirements, we identified and summarized the main features of the four most appropriate open-source IoT platforms namely ThingsBoard[3] , DeviceHive[4], Zetta[5], and SiteWhere[6]. A thorough comparison on the basics of key properties which are important for the platform selection is presented in Table. I.

Each platform has its own advantages and disadvantages, but it can be considered satisfactory depending on the specification of each case-by-case scenario. In a healthcare environment where it is necessary to integrate various heterogeneous network of sensors and where the bidirectional communication between the platform and devices is needed, the platforms must support multiple protocols and the capability of customizing integration to interact with devices and external systems in an untroubled manner. In addition, a degree of support from developer communities is a decisive factor that should be considered. From this perspective, ThingsBoard and SiteWhere seem to be ahead from the rest of the examined platforms. However, based on the re-engineering capability of ThingsBoard for multiple applications, we decided to make up from this open source as the core IoT platform of our system.

### B. The System Overview

An overview of the system is given in Fig. 4. Basically, the proposed architecture follows the IIRA three-tier design. The ThingsBoard platform was used for the implementation of both Platform and Enterprise tiers. (1) At the Edge tier, sensor devices collect and transmit data via HTTP, MQTT, CoAP and other protocols through an IoT Gateway to the central cloud server for data transformation and storage. (2) The Platform tier consists of the modules of ThingsBoard. Specifically, the built-in Rule Chain Engine is used for data transformation and basic data analysis, where the incoming and ongoing data is filtered or transformed following a rule-based logic. The data is forwarded to the ThingsBoard Core Services module for storing into the local database and to the Advanced Data Analytics Module for data processing, leveraging the platform's in-built a Queue Message Module. The Queue Message Module includes different stream

[3]https://thingsboard.io

[4]https://devicehive.com

[5]https://www.zettajs.org

[6]https://sitewhere.io



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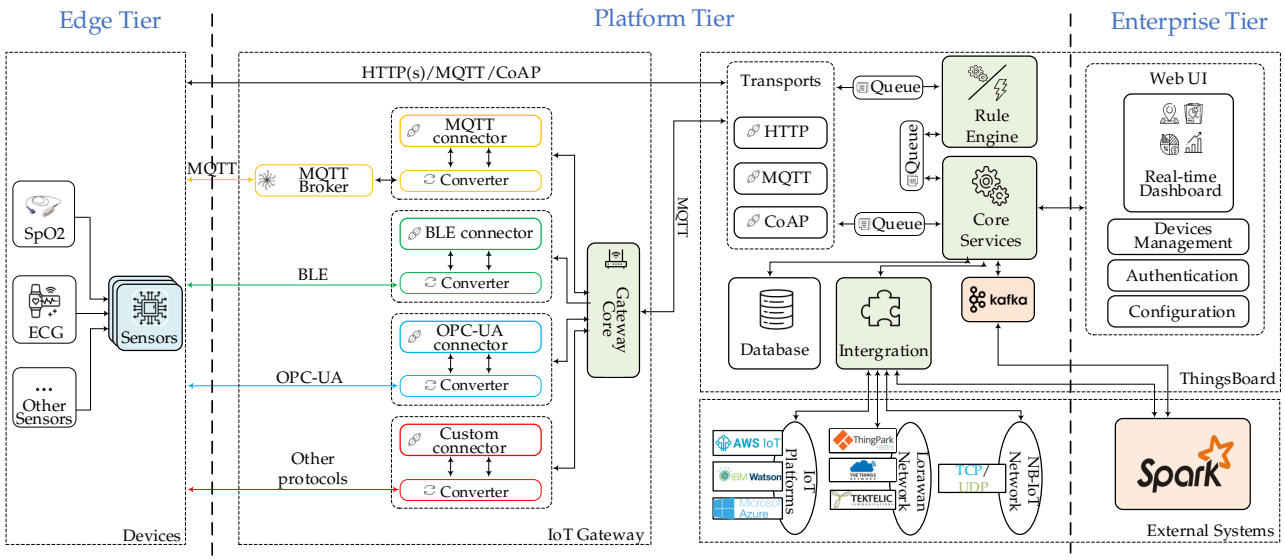


Fig. 4. The proposed system architecture based on IIRA three-tier model

management services such as Kafka, RabbitMQ, AWS SQS, and Google Pub/Sub. A NoSQL Cassandra database for storing telemetry and device configuration data. In addition, other SQL databases such as MySQL and PostgreSQL are easily configured via Rule Chain Engine. For real-time data processing, Kafka works as a bridge topic connecting ThingsBoard to Spark Streaming. The Data Analytics module based on Spark Streaming pulls these data from Kafka topics, organizes them into discrete streams and starts processing them. For data intensive or heavy workload applications, such modules can also run on a cluster of machines. Then, the analyzed results will be pushed back to the ThingsBoard dashboard. Finally, (3) at the Enterprise tier, a web interface is built allowing the creation of dashboards with visualization of the telemetry data with interactive charts. The dashboard function is flexible depending on different applications.

### C. IoT Interoperability Handing Strategy

*1) IoT Gateway:* For the concern of exchanging data with heterogeneous devices, the proposed system offers an IoT gateway as a software to translate dissimilar communication technologies (i.e., BLE, ZigBee and OPC-UA) or support IoT devices with programmable firmware to directly communicate with the central cloud by using a pool of standardized protocols (i.e., HTTP(s), CoAP and MQTT) which is located in Transports Module. Obviously, it is crucial that standardized protocols are established for all devices. However, the gateway solution has limitations when integrating a new device with a new communication protocol, thus the gateway interoperability solution should be salable. Therefore, a Custom Connector function is implemented to support creating connectors for the required protocols.

Network interoperability, which is the communication capability of two networks, can be achieved in two ways: either by having the two networks conform to a common protocol standard or by providing a gateway that translates between the two protocols. It is not feasible for two networks to confirm in advance a communication protocol, so the proposed system should support the most common standard protocols. Then having gateway solutions is essential to enhance the network interoperability.

*2) Integration and Data Converter Modules:* The Integration module is designed for two primary use cases in ThingsBoard: (1) Link existing networks and other networks with specific payload formats directly to the central platform. (2) Stream data from devices connected to existing IoT platforms such as AWS IoT, IBM Watson or Azure IoT to enable efficient data processing. At the moment ThingsBoard supports various integration protocols. Most popular are HTTP, MQTT, Kafka and OPC-UA. This platform also supports integration with specific LoRaWAN Network servers, Sigfox backend, various NB IoT devices using raw UDP and TCP integrations. AWS IoT, IBM Watson and Azure Event Hub allows users to subscribe to the data feed from devices via MQTT. This integration leverages the system playing a role as a cross-platform that helps to resolve the problem of Platform Interoperability.

Besides, the functionality of Integration module partially overlaps with the functionality of IoT Gateway in terms of communication protocols like HTTP and MQTT. However, there is a key difference between them that is the scalability. The IoT Gateway module is designed for local network deployments, which supports a small number of devices, while the Integration module is designed for server-to-server integration and high throughput (i.e., reflecting the platform interoperability.

The system supports JSON, binary and text as the syntactic payload formats for communication. As mentioned in the


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Semantic Interoperability session, the syntactic format is different depending on a vendor or even software version. Thus, the Data Converter is designed to convert and extract a subset of meaningful information out of the incoming message from external platforms, then the exchanged data can be consumed.

### D. Integrating Kafka and Spark for Big Data Analysis

Figure. 5 shows the procedure of real-time processing data by using Spark. Specifically, the ThingsBoard platform provides the ability to transfer outgoing streaming data to external systems. It can be configured via the platform's Rule Chain Engine. There are two ways to process collected data including batch mode (historical data) and streaming mode (incoming streaming data). For the streaming mode, data collected from devices or third-party platforms can be stored in topics, which are the core abstraction that Kafka provides for a stream of records. A Kafka topic plays a role as the producer publishes a stream of records, then a consumer like a Spark instance may get the data written on the topic and perform various data processing functions. As an example, Spark's MLlib can be used to configure, train and deploy machine learning based applications, which can offer classification services such as anomalies detection of the IoT system (e.g abnormal operating conditions, inappropriate system status). For the batch mode, Spark can directly access data from ThingsBoard's database, obtaining subsets of the data which can be used to train models on the Spark instance. At the end, the processed data is pushed back to the ThingsBoard platform using Kafka Integration, and the results is displayed on the Enterprise Tier Web UI.

Another advantage when adopting Spark platform for Big Data analysis is that Spark uses functional programming language, Scala, which is ideal for distributed systems. So that the data can be processed on a wide scale.

## IV. SpO2 Monitoring and Analysis Case Study

Monitoring the hypoxic condition in blood plays an important role in cancer diagnosis and treatment. In this case study, the proposed system has been deployed and implemented in order to examine new efficient approaches SpO2 acquisition and analysis, archiving the interoperability

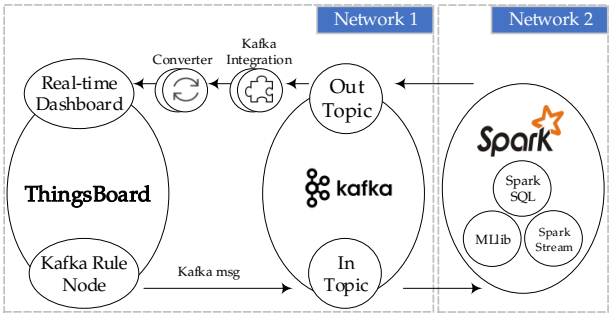


Fig. 5. Workflow of big data processing with Kafka and Spark platforms

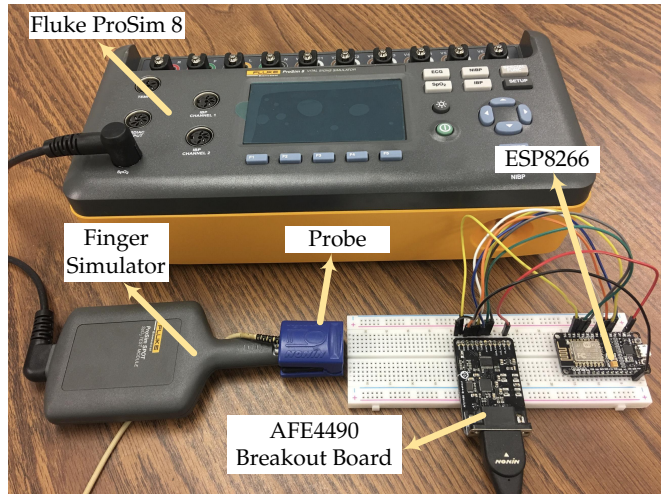


Fig. 6. Setup at the edge tier for the SpO2 data acquisition case study

and the ability of big data processing. For the data analysis, we performed a simple peak detection algorithm on the collected SpO2 signal in a Spark instance. After that, we calculate the heart beat rate and finally show the results back to the dashboard. This is an IoT demonstration of our previously developed system for the monitoring and diagnosis of obstructive sleep apnea [52].

### A. System Deployment

The deployment can be divided into two parts including hardware and software components. The hardware prototype is described in the Fig. 6.

*1) Hardware Components:*

- Sensor and analog-front-end design: photoplethysmography (PPG) sensor from a Nonin SpO2 probe and analog signal conditioning circuit have been connected in a breakout board. The PPG sensor measures the volumetric variation of blood circulation in the microvascular bed of a tissue to estimate the blood oxygen level. The measurement is done by a standard commercial probe (Nonin) that clips on to a simulating human finger which contains emitters as well as a light sensor. The AFE4490 IC to collect analog signals from the optical sensors and convert to digital to feed in the microcontroller.
- Microcontroller (MCU): The ESP8266, a low-cost wireless MCU built by Espressif [53] has been utilized for the data acquisition and transmission. It plays a role as the central unit to get and send the sensors data over WiFi to ThingsBoard.
- Signal generator: The Fluke ProSim signal generator has been used to generate the simulated data for the testing and validation of the proposed system. This machine can generate simulated SpO2 signals in different parameters corresponding to different health conditions via finger simulators connected to the SpO2 sensor probe.


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*2) Software Components:* The IoT Gateway, ThingsBoard and Spark software are installed on virtual machines created and managed by the Center for Computationally Assisted Science and Technology (CCAST)[7] at North Dakota State University, Fargo, USA. In particular, ThingsBoard plays a role as the brain for SpO2 data collection and visualization. The IoT Gateway is responsible for routing data using appropriate protocols, and the sensor-collected data will be analyzed by using the Spark platform.

### *B. Data Acquisition and Analysis*

Fig. 7 demonstrates the workflow of data collection and analysis among entities within the system. At first, the SpO2 data generated from the Fluke simulator (i.e., simulated patient) is collected by the AFE4490 IC on the ESP8266 MCU module. Next, the ESP8266 MCU, which is programmed and configured as an MQTT Client, receives, processes and publishes the sensor-collected data using the MQTT protocol. The data is transmitted the IoT Gateway, which is configured as an MQTT broker. After that, ThingsBoard receives the data as telemetry messages from the IoT Gateway and displays it on the dashboard, as shown in Fig. 8. To demonstrate the capability of data analysis, a SpO2 peak detection algorithm is implemented in a Spark instance to estimate the heart rate from the streaming SpO2 data. Finally, the Kafka integration is configured to display the heart rate calculated from Spark on the dashboard. The data transmission and analysis using HTTP protocol have also been implemented similarly.

### *C. Evaluation and Experimental Results*

*1) Data Transmission Evaluation:* In order to evaluate the transmission efficiency between the Edge and Platform tiers, the data loss and latency of the raw SpO2 data generated from the prototype devices and that received on the cloud server were collected. Specifically, we carried out 10 trials with SpO2 data in 3 different health conditions including normal, hypertension and hypotension conditions, each trial in 30 seconds with the sampling rate of 25 Hz. Packages loss and latency metrics are used to evaluate the experiments. Assuming

[7]https://www.ccast.ndsu.edu

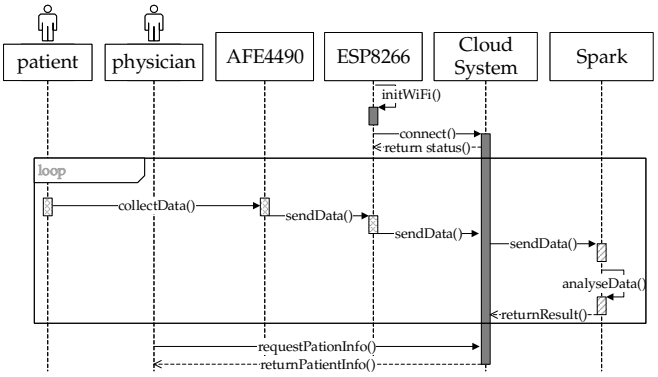


Fig. 7. Sequence diagram of SpO2 acquisition and analysis case study

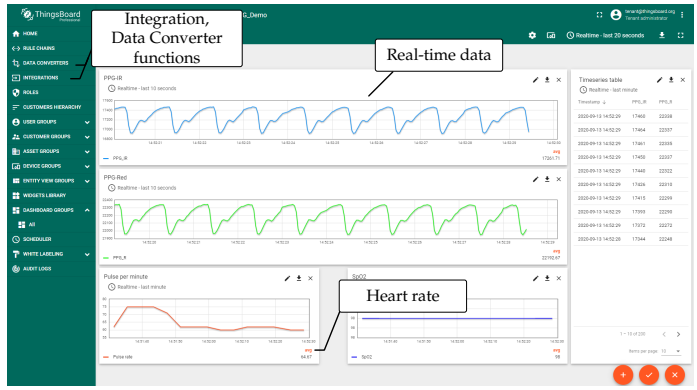


Fig. 8. Demonstrative dashboard of the SpO2 monitoring and analysis case study

TABLE II
THE MEASUREMENT OF DATA LOSS AND LATENCY OF SPO2 TRANSMISSION FROM THE EDGE TO THE PLATFORM TIER WITHIN 30 SECONDS. THE SAMPLING RATE IS 25 HZ.

| **Num of Trials** | **Data Loss** (%) $\bar{x} \pm s$ | **Latency** ($ms$) $\bar{x} \pm s$ |
|---|---|---|
| 30 | $0.1231 \pm 0.3061$ | $631 \pm 386$ |

that the network performance will remain the same over short period of testing time and the simulator can generate the exact same signal with same configuration, the mean and standard deviation of packages loss rate is $0.123 \pm 0.0361$ and latency is $631ms \pm 386ms$ as shown in Table. II.

The standard deviation of packages loss rate and latency indicate that the amount of variability from the individual data values to the mean is pretty low when transmitting data from the Edge to the Platform Tiers.

*2) Performance evaluation of gateways and data converter:* In order to evaluate the performance of IoT gateways and data converters, we measured metrics on response time when sending identical payload with different formats such as JSON and text through MQTT and HTTP protocols. Specifically, the end device (i.e., AFE4490 sensor) is simulated on a laptop, which runs both an MQTT client and an HTTP client, and then we measure the response time for sending a single message to the cloud system with 1, 10, and 100 property fields over a single connection cycle each (i.e., variation in payload size) over the Internet by using Wireshark[8].

As shown in Fig. 9, we could note that response times decrease dramatically as the size of the payload increases with the use of MQTT compared to HTTP. Another observation is that with the payload as binary format, the response times is higher than that payload as JSON format. It can be explained as the binary format encodes the data in a more concise description with the trade-off of low human readability.

*3) Data analysis using Spark:* The incoming SpO2 data from Kafka streaming is ingested and stored in the setup Spark

[8]https://www.wireshark.org


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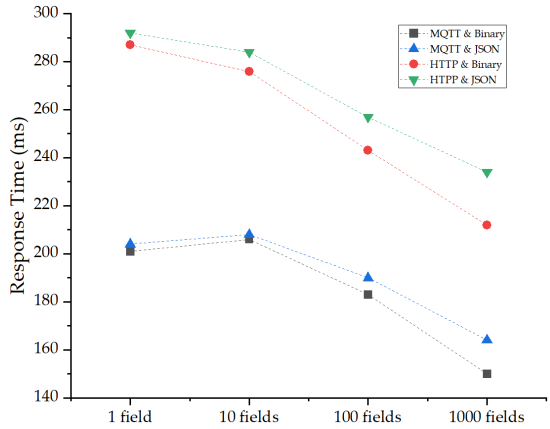


Fig. 9. Comparison of the response times in sending a single message from Edge to Cloud System with different payload sizes (1, 10, 100, and 1000) and payload formats (JSON and Binary)

instance. Using Spark APIs, we performed a simple peak detection algorithm [54] on the collected data. A representative of 20 second signal with detected and rejected peaks is shown in Fig. 10. The computed heart rate is pushed back to the ThingsBoard dashboard for data visualization and real-time monitoring purposes.

## V. Conclusion and Future Work

This paper proposed a skeletal IoT infrastructure for smart connected health applications. The proposed system utilized the IoT gateway, integration and data converter solutions to enhance the IoT interoperability. The integration of open-source and analytic platforms has facilitated the real-time analysis of streaming data. In addition, the case study of the real-time SpO2 signal monitoring and analysis based on the skeletal IoT architecture has been investigated. Through this case study, we demonstrated that the implementation of reference IoT architectures using open-source platforms is feasible in enhancing the IoT interoperability and potential in opening opportunities related to scale applications and analysis.

Future work involves the deployment of big data analysis using Spark and Kafka integration in the micro-services scenario. The obtained data will be processed and analyzed in real-time within a distributed system to reduce the reliance on single-point failures and increase the system's robustness, responsiveness, and scalability. Besides, a plan for data bench-marking analysis has been scheduled. The SpO2 signal collected from the proposed system needs to be validated with the gold standard such as Polysomnography (PSG) system.

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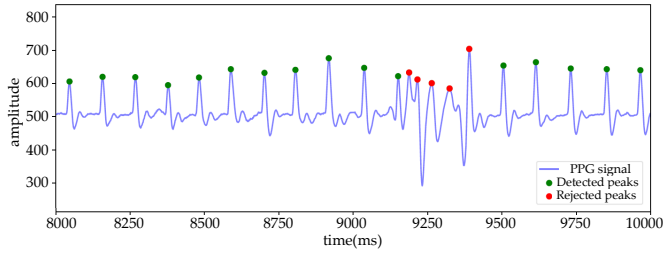


Fig. 10. Peak detection of a representative 2s segment of SpO2 signal sampled as 100Hz with the detected peaks and the rejected peaks

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